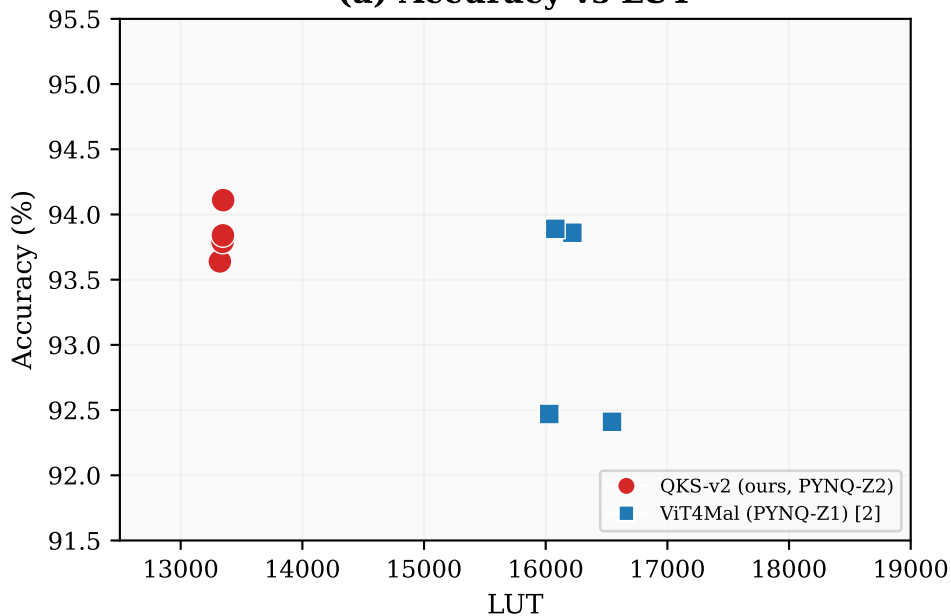
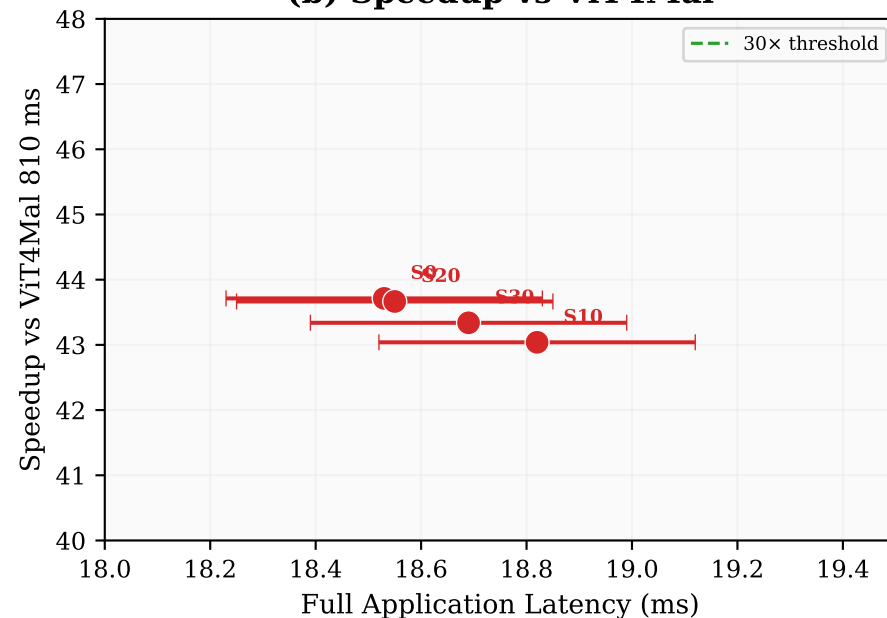


QKS-v2: Q/K Physically Pruned Structured Sparsity — PYNQ-Z2 (3,444 IMG_DS images)

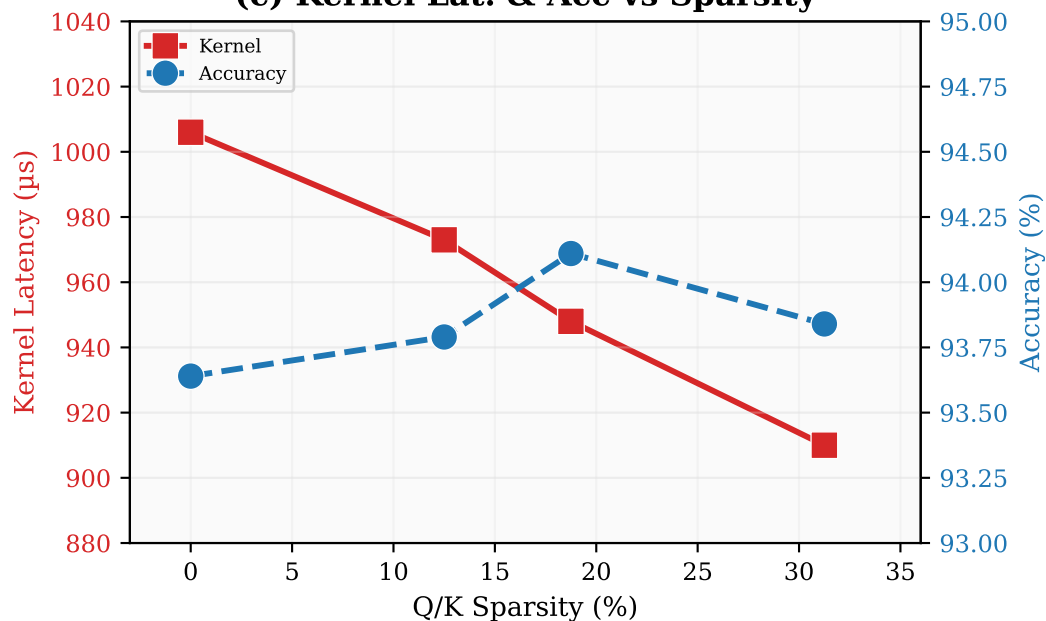
(a) Accuracy vs LUT



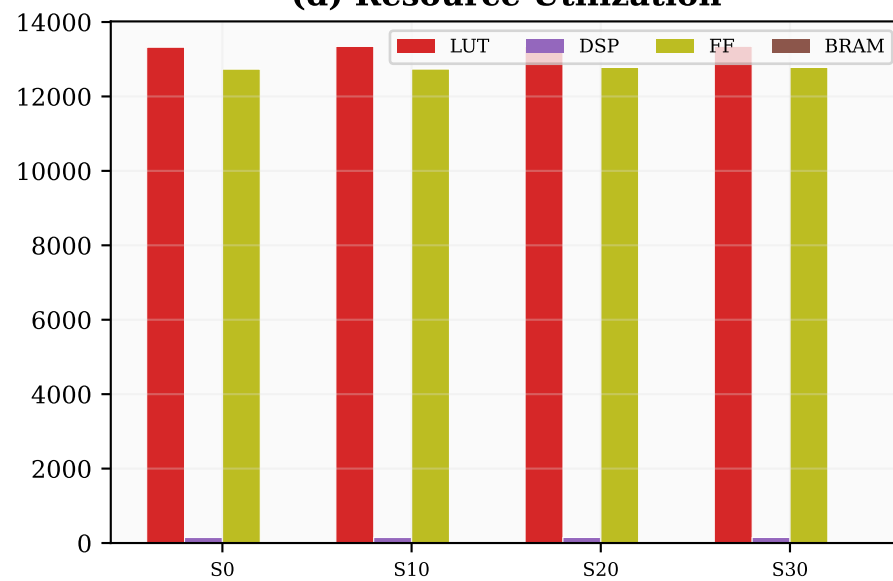
(b) Speedup vs ViT4Mal



(c) Kernel Lat. & Acc vs Sparsity



(d) Resource Utilization



Q/K physical pruning: weights $W_q/W_k[ATTN_DIM \times 16]$ (not mask). HLS loop bounds verified. Kernel: 1006→910 μ s (−9.6%). Acc: 93.6–94.1%. Speedup: 43.1–43.7 $\times$ (conservative claim >30 $\times$).